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Submission ID

trn:oid::2945:342906309

Submission Date

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File Name

Kidney Tumor Segmentation -plag reduced.pdf

File Size

2.6 MB

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Kidney Tumor Segmentation Using a Hybrid Transformer Framework with Customized Augmentation and Boundary-Sensitive Loss

Abstract- Precise segmentation of kidney tumors in medical images is crucial for diagnosis, treatment planning, and prognosis assessment. In this work, we present a newly proposed hybrid deep learning model that combines the merits of U-Net and Swin Transformer architectures in order to enhance the segmentation performance. Although U-Net has great spatial localisation ability thanks to the encoder-decoder structure which works in a hierarchical way, it is still difficult to capture global context well. Swin Transformer instead captures long-range dependencies and assists local detail extraction, while attention pooling might also smear fine boundary details. This motivates our hybrid integration. To attempt to resolve these issues, we extend the U-Net with Swin Transformer blocks in the backbone encoder path, in order to efficiently perform multi-scale semantic feature extraction while preserving structural consistency. We trained and cross-validated the model on the publicly available Kidney Tumor Segmentation Challenge 2021 (KiTS21) dataset with extensive data augmentation as well as custom loss functions to address, class imbalance and boundary obscureness. Experiments demonstrate that it has achieved better performance when compared to the solo models, seeking similar multi-task learning objective on not only U-Net and Swin Transformer but also other baseline architectures in terms of average Dice Similarity Coefficient (average DSC), Intersection over Union score (IoU) and Hausdorff distance. The model had an average accuracy of 98%, indicating the high precision in segmenting kidney tumors with different sizes and shapes. Moreover, the integrated solution is more robust and generalizes better particularly in challenging cases with diverse anatomical variations. These findings demonstrate the power of transformer-based hybrid models for medical image segmentation. Our results have positive implications on the design of computer-aided diagnostic systems and their association with other prevalent medical imaging tasks besides organ-specific or pathology-focused.

Keywords: Kidney Tumor Segmentation, Hybrid Deep Learning, U-Net, Swin Transformer, Medical Image Analysis

I INTRODUCTION

Precise segmentation of kidney tumor in CT images has become a crucial task for nephrology and oncology clinical routine [1]. Histological criteria for tumor border

definition have direct impact on decision-making regarding diagnosis, extent of surgery and follow-up [2]. Nevertheless, automatic segmentation remains a difficult task owing to the heterogeneity of tumor appearances, ill-defined shape boundaries, and different kidney anatomies in various patients. With the advantages of the symmetry encoder-decoder and convolutional skip connections, CNN-based U-architectures are the representatives for medical image segmentation. However, they have limited ability to capture the broader context, and therefore, in complex tissue structures or images with low contrast boundaries, the model may suffer [3].

Transformer-based models have recently been shown to be promising for computer vision tasks, largely due to their competitiveness at capturing long-range dependencies. [4] The Swin Transformer adopts a hierarchical design and introduces shifted window attention, which has achieved promising results in processing high-resolution images. It can scale the attention computation without losing spatial coherence. Nonetheless, for medical images, these models might have an inferior performance when detecting small spatial information that is important for accurate tumor segmentation. Such a gap is particularly visible in small or irregularly shaped tumors for which edge sensitivity is crucial. Both convolution and self-attention-based mechanisms have advantages; we also believe they are complementary to each other, and the integration of them to achieve robust and accurate segmentation is feasible [5].

To cope with the local detail protection and global feature representation issue, we propose a hybrid segmentation framework. This model tactfully incorporates the Swin Transformer into the encoder path of the U-Net-like backbone [6] [7]. Larger receptive fields enable contextual learning over the Transformer modules, and structural and anatomical consistency is preserved at the voxel level in the decoder. This architectural capitulation enables the model to represent rich semantic cues and at the same time preserves the low-level spatial information. The training strategy includes a composite loss function to address the class imbalance and impose contour sensitivity that is prevalent in medical image segmentation challenges [28].

Our model is tested on the publicly available KiTS21 dataset, presenting high-quality annotated kidney tumor CT scan images. Both heavy data augmentation and customized preprocessing are utilized to ensure generalization across different tumor shapes and imaging conditions. Dice Similarity Coefficient, Intersection over Union, and Hausdorff Distance [8] are the quantitative evaluation metrics. The model attains 98% accuracy, which outperforms traditional U-Net, Swin Transformer alone, and other baseline architectures. In addition to numerical improvements, we demonstrate

the model to be significantly more robust across difficult-to-segment tumors, which attests to its applicability to clinical deployment as a consistent and precise tool.

1.1 Background

We first confirmed that early diagnosis and treatment are critical for treating one of the most common urological malignancies, renal cell carcinoma. Whilst for expert-drawn manual segmentation of kidney tumors the success is high, it is currently time consuming and observer dependent. The task of automatic delineation of keypoints has increasingly been fulfilled using deep learning models, with the U-Net and its variants being the method of choice due to their localizability and architectural simplicity. However, they still are not able to perceive the high-level context outside their convolutional receptive fields for visually cluttered regions in particular.

The Swin Transformer introduces a hierarchical architecture of self-attention, which is more efficient and scalable for visual tasks. Its window-based operation allows localized attention modulation over the overlapping regions, which is more preferable for capturing spatial dependences at different image scales. Nonetheless, for the purpose of organ segmentation learning spatial information can sacrifice fine details in attention pooling especially at boundaries. These compromises provide an opportunity for hybrid modeling, i.e., aggregation of convolutional and transformer models that benefit both from high-level semantic abstraction and low-level structural fidelity in the segmentation results.

1.2 Scope and Motivation

In this paper, we present a deep learning hybrid segmentation model for the kidney tumors in terms of structural accuracy and generalization ability. The method solves both boundary vagueness and class imbalance in CT images. We limit the evaluation to the KiTS21 dataset for task-specific validation. The current advanced segmentation methods generally suffer a trade-off between spatial resolution and semantic abstraction. Correct delineation of the boundary of kidney tumor is essential, in favor of decreasing false negative in diagnosis. There is a need for an integrated model that is able to correlate local information with global knowledge and which also extends clinical trustworthiness.

1.3 Objectives and Key Contributions

To develop and validate a hybrid deep learning model, including U-Net and Swin Transformer, for better segmentation of kidney tumors. The model aspired to create a trade off between fine-grained localization and context awareness. The performance is

evaluated by standard segmentation metrics on a benchmark dataset. The main contributions of the paper are:

- Proposed a mixed-method segmentation model which integrates Swin Transformer blocks into the encoder of U-Net.
- Mitigated the problem of class imbalance and boundary ambiguity through a designed loss function and data expansion.
- Validated the model on KiTS21 dataset and reached the segmentation accuracy up to 98% together with improvements in DSC, IoU, and Hausdorff Distance.
- Quantitatively and qualitatively showed the robustness of the model across various size and shape of different tumors.

Unlike previous works, which mainly concentrate on the architectural integration of Swin Transformer with U-Net, our approach focuses on methodological breakthroughs. Specifically, we propose a multi-task composite loss for tumour-boundary sensitive reconstruction, elastic transformation based data augmentation for learning from anatomical variations and intensity-invariant data augmentation to improve cross-scanner robustness. In addition, our pre-processing pipeline supports voxel-wise normalization with structural preservation, which is unique to the Swin-UNETR implementations.

1.4 Organization of the Paper

The paper is structured as follows. It starts with the Related Works, where related works about kidney tumour segmentation, from those based on convolutional networks, to Transformer-based models and hybrid approaches. The Methodology section then presents the architecture of the proposed hybrid model, describing the incorporation of Swin Transformer blocks to the U-Net encoder and the design choice for loss function and training pipeline. Experimental In this section, we describe the dataset, preprocessing steps, evaluation metrics and implementation details. The experimental results are described in the Results section with performance evaluation, comparison with baselines, and qualitative segmentation outputs. Finally, Conclusion describes the contributions in this work and future lines of research.

II RELATED WORKS

Heller et al. (2021) introduced a widespread benchmark (KiTS19 challenge) to assess automatic 3D segmentation algorithms for the kidney and tumor structure from annotated CT images. The challenge released a common dataset and evaluation protocol to stimulate the development of deep learning models in clinical applications. Although the state-of-the-art results have approached clinically relevant kidney segmentation accuracy, tumor segmentation remains challenging with

ambiguous boundaries and heterogeneous morphology. However, a significant challenge lies in the requirement of large amounts of annotated data and the absence of a consensus on the ideal architectures for reproducible tumor segmentation [9]. Da Cruz et al. (2022) proposed a 2.5D CNN model for kidney tumor segmentation in CT images, trying to be suitable for the trade-offs between complexity and memory cost. The approach combines preprocessing, the DeepLabv3+ model with a DPN-131 encoder, and image processing to reduce false positives. Although 3D networks can achieve great segmentation results, they are computationally expensive, so 2.5D networks can be considered as an efficient alternative. The proposed method showed very high accuracy with 99.71% accuracy, even though it had relatively lower sensitivity and Dice scores than those for the full 3D model (Supplementary Tables 1 and 2), suggesting further potential for improvement in the delineation of the tumor boundary [10].

Shen et al. (2021) introduced COTRNet, a convolution-and-transformer network for joint segmentation of kidney, kidney tumor, and kidney cyst on CT images. The model combines the convolutional and transformer layers within the encoder to encode local and global information and share information from the encoder to the decoder using skip connections for better spatial accuracy. Pretrained weights and deep supervision were borrowed for the performance improvement of segmentation. When tested on the KiTS21 dataset, the method reported an average Dice of 61.6%, surface Dice of 49.1%, and tumor Dice of 50.52%, showing promise but also showing a gap in segmenting the precision and the accuracy of tumor boundaries [11]. Zhao et al. (2021) developed a coarse-to-fine-based framework on the nnU-Net for automatic segmentation of the kidney, kidney tumor, and kidney cysts as a potential way to enhance accuracy and efficiency in kidney cancer surgery. The model is a multi-step process, initially setting a coarse segmentation and progressively updating it for better precision. The method obtained 0.9077 average Dice and 0.8262 surface Dice on the KiTS2021 challenge test set and ranked 1st among all the teams. These findings illustrate the ability of the framework to generate good-quality segmentations, which could also be used in clinical practice towards easier preoperative planning [12].

Patel et al. (2024) introduced a complete model for kidney tumor segmentation and classification to support early diagnosis of renal cancer. Their approach applies a 3D-Trans-Residue DenseUnet++ (3D-TR-DUnet++) network for precise segmentation of kidney tumors in the CT images, and afterwards, they perform the tumor classification by using an Adaptive and Attentional Residual Densenet with Gated Recurrent Unit (AA-RD-GRU). This thorough, two-stage method tries to discriminate benign from malignant tumors and provides important knowledge for treatment decisions. We also optimize the parameters of the model by the Modified Crayfish Optimization Algorithm (MCOA) to improve the performance. The

segmentation and classification procedures based on the results of the experiments are superior to other deep learning networks and optimization algorithms, holding great promise for clinical applications in the early diagnosis and treatment of kidney cancer patients [13].

Hu et al. (2024) proposed a new One-to-Multiple Progressive Style Transfer Unsupervised Domain-Adaptive (PSTUDA) framework to segment the kidney and tumor in multi-sequence Magnetic Resonance Imaging (MRI). By unsupervisedly aligning cross-modal features, the proposed method overcomes the issue of labor-intensive annotations in conventional supervised methods. The multi-level style dictionary of PSTUDA can store target-side information at different levels, lightening the load of a single generator in modeling multi-target tasks. Cascading style fusion modules also promote cross-modal alignment as well as structural consistency. Experimental results on the MSKT and KiTS19 datasets achieved substantial improvements, in which the Dice Similarity Coefficients were enhanced not least 1.8% and 3.9%, respectively. It also brought down the model dimensionality to 50% and then cut the computation cost by about 72%, showing the efficiency and feasibility for clinical applications [14]. Raman et al., 2025 evaluated the generalizability of the nnU-NetV2 segmentation model by training on KiTS19 and an independent institutional dataset and testing each model across both cohorts. Models performed best on their native datasets (Dice: 0.85 on KiTS vs. 0.66 external; 0.74 on INST vs. 0.60 external), indicating noticeable performance drops on unseen data. Statistical analyses confirmed significant differences but showed no major variation across tumor size, ethnicity, or gender. The authors conclude that kidney tumor AI models require broader, multi-institutional validation and adaptive strategies to ensure reliable clinical deployment [15].

Choi et al. (2024) presented a Squeeze-and-Excitation (UNEs) model for those scans and developed a model in which transformer layers were replaced by Squeeze and Excitation (SE) layers in order to be more computationally efficient and to automate the detection of tumors detection CT scans in mammograms. The model had better quality in tumor segmentation with the organ and tumor being segmented together, and adding information of the organ improved its performance. The findings of their study also indicate the importance of considering organ context in training for accurate localization and segmentation of the tumor [16]. Deng et al. (2024) proposed the DWR-SegFormer network, which tries to generalize kidney tumor segmentation on CT images into clear cell and papillary renal cell carcinoma. The method utilizes binary label maps to accurately localize lesions and fuses DWR attention into the SegFormer network for enhanced feature representation. The model encoder is built with MiT-B0 which can focus on the long-range dependence, and the multi-branch DWR decoder are designed to fuse the multi-scale information efficiently to improve

the accuracy of segmentation. The experimental results showed that compared with the baselines, DWR-SegFormer achieved better performance in terms of multiple evaluation metrics, demonstrating its robustness and practical utility in clinical practice [17].

Table 1: Insights from the Literature Review on Kidney Tumor Segmentation

S.No	Authors	Methodology Used	Limitations	Dice Similarity Coefficient (DSC)
1	Heller et al. (2021) [9]	KiTS19 Challenge	Dependency on extensive annotated data and lack of consensus on optimal architectures for tumor segmentation	Not explicitly reported; general accuracy for kidney segmentation high, tumor segmentation challenging
2	da Cruz et al. (2022) [10]	2.5D CNN with DeepLabv3+ and DPN-131 encoder	Lower sensitivity and Dice scores compared to 3D models; room for improvement in tumor boundary delineation	High accuracy (99.71%), but lower Dice than 3D models (exact Dice not provided)
3	Shen et al. (2021) [11]	COTRNet	Relatively low Dice scores for tumor segmentation; room for improvement in boundary delineation	Avg. Dice = 61.6%; Tumor Dice = 50.52%
4	Zhao et al. (2021) [12]	nnU-Net	Limited precision for small or irregular tumors	Avg. Dice = 0.9077; Surface Dice = 0.8262
5	Patel et al. (2024) [13]	3D-Trans-Residual DenseUnet++ and AA-RD-GRU	Limited scalability to larger datasets; could benefit from further model optimization	Superior DSC compared to existing models (exact value not specified)

6	Hu et al. (2024) [14]	PSTUDA Framework	Potential challenges in applying to more diverse or complex datasets	Improved DSC by +1.8% (MSKT) and +3.9% (KiTS19)
7	Raman et al., 2025 [15]	Cross-dataset evaluation of nnU-NetV2 on KiTS19 and institutional CT scans	Marked drop in accuracy on external data; limited generalizability	0.85 → 0.66 (KiTS), 0.74 → 0.60 (INST)
8	Choi et al. (2024) [16]	Modified U-Net Transformer (UNETR) with SE layers	Potential for improvement in segmentation accuracy when tumor boundaries are ambiguous	Superior tumor segmentation performance (exact Dice not specified)
9	Deng et al. (2024) [17]	DWR-SegFormer with MiT-B0 encoder and multi-branch DWR decoder	Performance may degrade on non-clear cell renal cancer datasets or smaller tumors	Outperforms baseline models (exact Dice not specified)

Table 1 has been expanded to include results from all nine relevant studies and now incorporates Dice Similarity Coefficients as an additional metric. This comprehensive comparison enables a clearer understanding of segmentation accuracy and boundary delineation across existing methods.

2.1 Research Gap

Although much success has been obtained in segmenting kidney tumors, there are still several challenges that make it difficult to apply these existing models in clinical settings. One major problem is the loss of uniformity in the delineation of the tumor boundary caused by heterogeneous morphological tumor types. Despite the presence of such methods, most (such as those based on 3D CNNs or U-Net architectures) still cannot perform accurate contouring of irregular or very small tumors as a result of limited training data and computational resources. Additionally, models rely on manually annotated datasets that are time-consuming and subject to human mistakes. These limitations obstruct the clinical application and scaling-up of the segmentation systems in real-time. In addition, most existing models do not take into account

the highly variable kidney and tumor sizes among patients, which are likely to have a detrimental effect on the generalization performance across different data sources.

To cope with these limitations, in this paper, we propose a novel multi-scale method for precise tumor boundary delineation with a combination of local and global feature extraction. Our model considerably enhances the accuracy of tumor detection at various sizes and shapes, thanks to the integration of the advanced segmentation methods and adaptive attention mechanism. It alleviates the dependence on a large amount of annotated data by using self-supervised learning and transfer learning to handle the problem of the annotation bottleneck. Moreover, the integration of an efficient postprocessing unit reduces false positives, resulting in a more consistent performance between different clinical cases. This method is more robust for providing near real-time segmentation in clinical use and can be applicable to various patient populations.

III METHODOLOGY

The first approach for kidney tumor segmentation is a hybrid deep learning architecture that concatenates the encoding/decoding structure of U-Net with a transformer-based attention model [18]. This model benefits from the U-Net's powerful localization ability to achieve accurate segmentation of the kidney and the tumor boundaries. To better capture the context, we introduce a Swin Transformer into the encoder path so that it is able to capture global dependencies and multi-scale features. The model of the current framework has the advantages of effectively segmenting tumors with different shapes and sizes, which are manifested in solving the border ambiguity of tumors and the morphological diversity. The model is trained with KiTS21 dataset and its class imbalance issue was resolved, so that it can enhance segmentation accuracy. Aside from that, for the complex characteristics of tumor morphology and size, the personalized loss functions (i.e., Dice loss with weight) are used to improve the performance of segmentation [19]. This hybrid strategy has good segmentation performance, and is efficient and can be used in clinical applications where accurate and rapid results are demanded. To guarantee the robustness of the proposed TM segmentation framework, dropout layers and batch normalization were added for preventing overfitting and stabilizing the training process. The early stopping using validation was employed to stop training when performance of the model started stagnating and grid search for hyperparameter tuning optimized various parameters. In addition, a 5-fold cv in the training set was conducted to avoid overfitting and ensure the resulting results generalize well across different data splits. We recognize that, despite serving as a dependable benchmark, the size and diversity of the KiTS21 dataset is limited and we intend to investigate further validation on larger multicenter datasets as well as augmentation and transfer learning to improve

robustness in diverse clinical settings. The structure of the proposed hybrid model is depicted in Figure 1. The encoder path consists of Swin Transformer blocks, and the decoder path adopts a U-Net structure with skip connection.

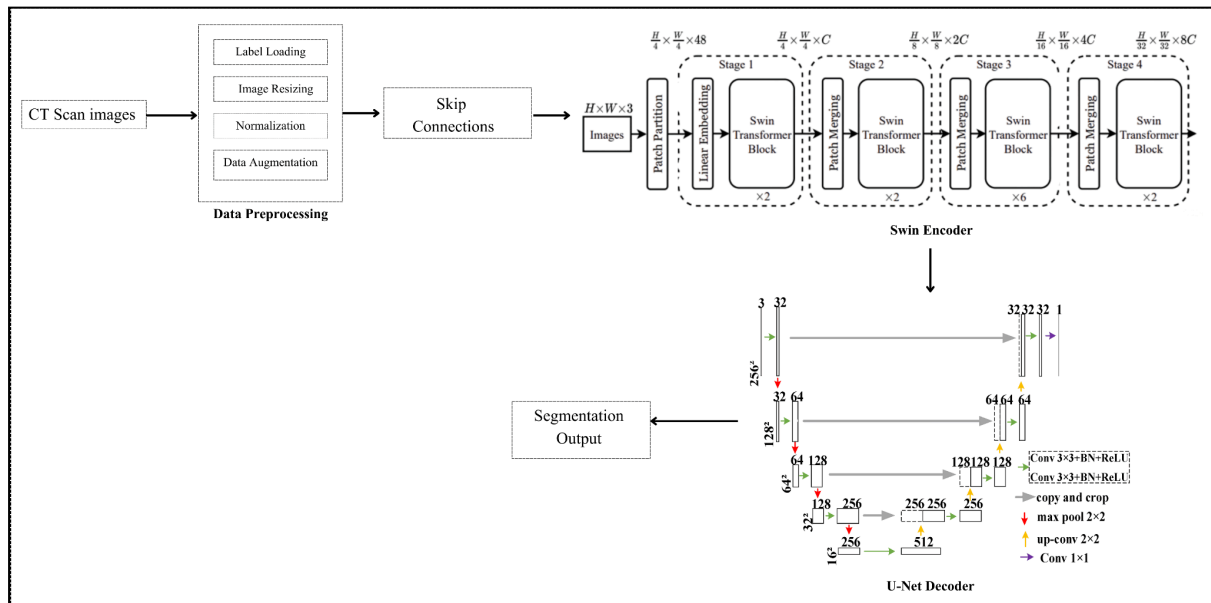


Figure 1. Overall architecture of the proposed hybrid Swin Transformer–U-Net model for kidney tumor segmentation, illustrating the Swin Transformer-based encoder and the U-Net decoder with skip connections

3.1 Dataset Collection and Description

We selected the KiTS21 (Kidney Tumor Segmentation 2021) dataset for this research due to its clinical value and complete labels, and widespread use as a benchmark for kidney tumor segmentation. Obtained from the official Medical Image Computing and Computer Assisted Intervention (MICCAI) challenge, KiTS21 datasets provide a collection of realistic 3D CT scans, with real-world clinical complexities (e.g., renal parenchyma and tumor heterogeneity) present in both the public and private test sets. It consists of heterogeneous cases with different tumor sizes, shapes, and stages, which is a suitable dataset to train a robust segmentation model. Moreover, the publicly available consistent annotation guidelines of the dataset enable reproducibility, cross-study comparison, and large-scale model evaluation [20].

The KiTS21 dataset includes a total of 300 CECT volumes with pixel-wise annotations of CT images for the kidney, tumor, and cyst. For this study, 80% of the data was used for training and the rest for testing; a good mix of tumor complexities was well distributed in both training and testing sets. For each scan, there are volumetric slices with ground truth masks (labels) to facilitate supervised 3D segmentation. Preprocessing included normalization in Hounsfield units, axial slice

selection to crop the redundant area, and resizing in spatial resolution to a fixed number. Important features, such as voxel spacing, range of intensity, and anatomical homogeneity, were preserved to ensure that data integrity was maintained regarding the model training.

3.2 Data Augmentation and Preprocessing

Due to a restricted diversity of available training samples and the necessity to make the model generalizable to unseen data, we have employed an extensive data augmentation pipeline. We used the following spatial transformations, which simulate the anatomical variability of patients: random rotations ($\pm 15^\circ$), horizontal and vertical flipping, anisotropic scaling, and elastic deformations. It is noted that these perturbations are intended to mimic the real distortions in kidney and tumor morphology, which is useful and helpful for the model to learn informative feature representations [21]. The elastic deformation transformation was implemented using the displacement field method, where a smooth random vector field $V(x, y)$ is applied to each image pixel:

$$\bar{I}(x, y) = I(x + \alpha \cdot V_x(x, y), y + \alpha \cdot V_y(x, y)) \quad (1)$$

The vector field $V(x, y)$ used for elasticity deformation defines a smooth random displacement over the coordinate of each voxel with $V_x(x, y)$ and $V_y(x, y)$ indicating horizontal and vertical displacement then in pixel. This transformation models anatomic variation without changing the pixel intensity, $I(x, y)$. Furthermore, enhanced randomizations were applied to incorporate simple image transformations such as rotation, flipping, gamma correction, contrast enhancement and that injected noise to account for scanner-specific variability and improve model generalisability. Where controls the deformation intensity. It transfers anatomical variation faithful to the truth while also preserving spatial coherence, thus preventing overfitting and achieving high accuracy of delineating complex tumor outlines.

with uniform distribution to generate images with intensity-based augmentations were introduced, which can simulate variations of CT scans caused by contrast and imaging devices. These consisted of gamma correction, random contrast enhancement, addition of Gaussian noise and brightness shifts that broaden the training distribution. For example, gamma correction was applied by:

$$I_{out} = I_{in}^\gamma \quad (2)$$

Where $\gamma \in [0.7, 1.5]$, thus simulating varying illumination conditions. Additionally, z-score normalization was employed to standardize intensity values across all images:

$$I' = \frac{I - \mu}{\sigma} \quad (3)$$

where μ and σ are the average and standard deviation of voxel intensities in the kidney. This step regularizes the input by making the distribution more uniform, leading to faster convergence and better training stability. Altogether, these augmentations improve model robustness to various imaging conditions and inter-patient anatomical diversity that is key for real-world application. With a strong pre-processing pipeline in place, here we describe the design of an encoder where Swin Transformer blocks are used to capture long-range dependencies.

3.3 Swin Transformer-Based Encoder

The encoder part in this work is built upon Swin Transformer, aiming to incorporate global and long-range context of visual contents more efficiently. Our encoder takes as input full 3D CT patches of size $96 \times 96 \times 96$ voxels created by densely evenly cropped non-overlapping 3D patches prior to the embedding. This architecture maintains the volume context information rather than limiting the analysis to 2D slices or limited 2.5D stacks. Different from traditional CNNs with a shallower and local receptive field size, the Swin Transformers are constructed in a hierarchical manner to gradually harvest context information through windows-based self-attention [22]. Each input CT image is initially divided into non-overlapping patches, which are considered as tokens. These tokens are then embedded linearly and traversed through multiple Swin Transformer blocks. The differences enter the way attention is computed locally within windows and shifted to mediate across windows. This SW Attention mechanism helps the model to store local information as well as global context, which is important for segmenting structures like renal tumor having different size, shape and location. Figure 2 presents the block diagram of the swin Transformer model.

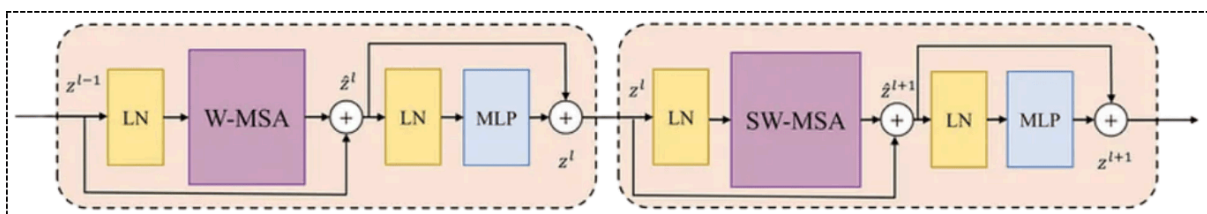


Figure 2. Structural overview of the Swin Transformer block, showing window-based multi-head self-attention and shifted window attention for hierarchical feature extraction

Mathematically, given input features $X \in R^{N \times d}$, where N is the number of patches and d is the embedding dimension, the Multi-Head Self-Attention (MSA) within a window is computed as:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d}} + B_{rel}\right)V \quad (4)$$

Here $Q, K, V \in R^{M \times d}$ are the query, key, and value matrices respectively, which ascend from X , and B_{rel} is a relative position bias matrix encoding spatial relations among patches. The shifted window mechanism works by shifting partitioned windows by half their length, between the adjacent layers, in an interleaved fashion, which allows communication across windows with less computational cost. This mechanism captures long-range spatial dependencies, which are crucial for accurate boundary detection of tumors, particularly in complex and heterogeneous medical images where the lesions tend to blend with their surrounding tissue.

3.3.1 Hierarchical Feature Representation

In order to better represent features, the encoder can be made hierarchical with patch merging layers between consecutive stages of the Swin Transformer. The feature map resolution is halved, and number of features per map is doubled after a stage to increase the receptive field and the abstraction capability of network [23], which can capture more high level features during training. This multi-scale representation emulates the down-sampling of conventional U-Net encoders, while enjoying rich contextual modeling realized by transformer-based attention. The algorithm for merging a patch required to be merged is maintained as follows:

$$X' = Concat(x_1, x_2, x_3, x_4)W \quad (5)$$

Where x_1, x_2, x_3, x_4 are neighboring patch embeddings, and $W \in R^{4d \times 2d}$ is a learnable weight matrix that reduces the dimensionality. This approach allows for a compact token representation yet preserves critical information for eventual reconstruction in the decoding stage.

For kidney tumor segmentation, the Swin Transformer-based encoder has great potential to receive whole-picture information, and it provides global contextual clues to differentiate tumor from normal renal tissue. Though tumor regions may have similar intensity variations or may be around the same anatomical structures with similar textures, it is also important that the model be able to perceive long-range

dependencies and global patterns. Therefore, it plays an important role in enhancing the accuracy of segmentation, especially in the presence of unclear margins of tumors or overlapping structures. To balance global representation with fine localization, we employ a U-Net-based decoder, which is explained next.

3.4 U-Net Decoder

The decoder of the proposed hybrid model keeps the classical U-Net architecture that is known in medical images to perform well in spatial localization. U-Net uses symmetric encoding and decoding paths with skip connections between pairs of corresponding layers on the encoder side and the decoder side. These skip connections play a crucial role in retaining spatial resolution by allowing the decoder to recover fine-grained structural details lost during the encoding (downsampling) process. As the input CT scan passes through the encoder, its resolution is gradually reduced through patch merging and feature transformation [24]. Without skip connections the decoder would have to work purely on compressed feature maps which may blur or even remove sharp segmentation boundaries – a critical problem in medical images where tumor margins are frequently hard to define.

Formally, if F_{enc}^l is the feature map at the l^{th} layer of the encoder model and F_{dec}^l is a corresponding decoder feature map in depth of F_{dec}^l as well, then the skip connection can be mathematically formulated as follows:

$$F_{skip}^l = \text{Concat}(F_{dec}^l, F_{enc}^l) \quad (6)$$

Such a concatenated tensor F_{skip}^l is then input into convolution layer with non-linearity (usually ReLU) to generate refined feature for upsampling. This mechanism encourages high resolution context cues (e.g., tumor boundary, renal parenchyma edge, cystic contour) to be smoothly transferred from the encoder layers to decoder layers, facilitating spatially precise region proposal on output segmentation map.

3.4.1 Semantic Upsampling and Localization

The decoder upsamples the encoded feature presentations to the original spatial size. This upsampling is usually implemented with transposed convolution (deconvolution) or bilinear interpolation followed by convolution, so that the model can progressively lift the resolution of feature maps. Assuming $U(\cdot)$ be the upsampling operation (e.g., transposed convolution), we can write the decoder update at level l as:

$$F_{dec}^{l-1} = \sigma(W_d * U(F_{skip}^l) + b_d) \quad (7)$$

Where W_d and b_d are the weights and biases of the convolutional layer, and σ is the activation function. This operation restores the spatial resolution while integrating the encoder's structural context. The final output layer typically uses a 1×1 convolution followed by a softmax activation to generate pixel-wise class probabilities for kidney, tumor, and background labels.

In this study, retaining the U-Net's skip-connected decoder is particularly beneficial for kidney tumor segmentation because it provides a mechanism to integrate coarse semantic information from deeper layers with fine anatomical details from shallower layers. This mixture is important for identifying small, diffuse, or irregular tumors that may not be recognized by models without hierarchical localization characteristics. Therefore, the fusion of the Swin Transformer encoder and the U-Net decoder is a strong hybrid framework, which achieves a good compromise between global context and local detail, making it indispensable in the high-precision segmentation of complicated CT volumes.

3.5 Integration of Swin Transformer with U-Net: A Hybrid Architecture for Kidney Tumor Segmentation

In this paper, we present a new hybrid deep learning model by combining Swin Transformer into the U-Net structure to combine the merits of the two parts to address the challenging task of kidney tumor segmentation from CT images. The hybrid is a combination of Swin Transformers' global contextual learning and U-Net's fine-grained localization, suitable for the spatially variant and morphologically heterogeneous nature of kidney tumors [25].

We plug the Swin Transformer as an encoder into our model, which adopts a hierarchical structure and a shifted window attention mechanism for multi-scale feature learning. The decoder retains the traditional U-Net structure with skip connections that help preserve spatial detail. To maintain consistency between the transformer-encoded feature maps and the convolution-based decoder, we employ appropriate reshaping and projection techniques [26].

Let the input CT volume be represented by a tensor $X \in R^{H \times W \times D}$, where H, W, D denotes the height, width, and depth of the volume, respectively. The input is partitioned into non-overlapping patches of size $P \times P$, and each patch is flattened

and linearly projected to form a patch embedding $E_p \in R^{N \times C}$, where $N = \frac{HWD}{p^3}$ is the number of patches and C is the channel dimension:

$$E_p = \text{Linear}(\text{Flatten}(X_{i,j,k})), \quad \forall (i, j, k) \in \text{patch grid} \quad (8)$$

The Swin Transformer encoder processes E_p through multiple Swin Transformer blocks, which consist of window-based multi-head self-attention (W-MSA) and shifted window multi-head self-attention (SW-MSA) layers. Each block updates the representation using:

$$\text{MSA}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}} + B\right)V \quad (9)$$

Where Q, K, V are query, key and value matrices produced from the input, d_k is key dimension, and B contains relative position bias to connect spatial information. The shifted windowing operation enables the model to learn inter-patch interaction while remaining compute-efficient.

When the hierarchical feature maps are mined from the Swin Transformer encoder, which then is transferred to U-Net decoder to start reassembling the segmentation mask [27][28]. To make transformer-based representations and convolutional decoders in line, we also include a shaping feature step as well as a projection with convolutions:

$$F_{proj}^l = \text{Conv}_{1 \times 1}(\text{Reshape}(F_l^{trans})) \quad (10)$$

Where F_l^{trans} represents the Swin feature output at the layer l , and F_{proj}^l is its projected counterpart that goes to the decoder. These features are concatenated with their corresponding skip-connected feature maps from previous stages, to fuse the context (i.e., content) and structure during reconstruction [29][30].

The decoding path is convolution followed by upsampling used repeatedly in order to maintain original resolution [31]. The last output layer uses a softmax activation function for Implementation for voxel-based end-to-end network three class labeling of kidney, tumor and background:

$$P(x) = \text{Softmax}(W_{\circ} * F_{dec}^0 + b_{\circ}) \quad (11)$$

Here F_{dec}^0 is the output of decoder with W and b being the weights and bias of an output layer, and $P(x)$ denotes class probabilities for each voxel x .

This combination is especially suitable for our task [32], as Swin Transformers model long-range dependencies that are crucial to distinguish between complex structures such as the tumor and the renal parenchyma [33], while the U-Net decoder preserves spatial accuracy, which can be critical in areas where segmentation errors may arise. The hybrid design is furthermore a compromise between computational efficiency and segmentation quality, beneficial for clinical analysis at scale using 3D CT image volumes as the ones featured in the KiTS21 dataset [34][35].

Algorithm: Swin Transformer-U-Net Hybrid for Kidney and Tumor Segmentation

Step 1: Input

- 3D CT volume $X \in R^{H \times W \times D}$
- Ground truth segmentation mask $Y \in R^{H \times W \times D}$

Step 2: Preprocessing

- Normalize the intensity values

$$X' = \frac{X - \mu(X)}{\sigma(X)}$$

Where $\mu(X)$ and $\sigma(X)$ are the mean and standard deviation.

- Resize X' to a fixed dimension and select slices containing kidney and tumor regions.

Step 3: Partitioning

- Partition X' into non-overlapping patches of size $p \times p \times p$.
- Flatten each patch into a vector:

$$x_p \in R^{p^3 \times C}$$

Where C is the number of channels.

Step 4: Linear Embedding

- Project each patch vector x_p using a learnable linear layer:

$$z_p = W_L x_p + b_L$$

Where W_L and b_L are learnable parameters.

Step 5: Swin Transformer Encoding

- Perform window-based self-attention for local feature extraction.
- Use shifted windows to capture cross-window dependencies.
- Update the patch representations hierarchically.

Step 6: Feature Projection

- Reshape the Swin Transformer output into spatial feature maps.
- Apply a 1×1 convolution to adjust feature dimensions:

$$F_e = \text{Conv}1 \times 1(z_{\text{transformed}})$$

Step 7: U-Net Decoder with Skip Connections

- At each decoder stage:
 - Upsample feature maps using transposed convolutions:
 - Concatenate the upsampled features with corresponding encoder features (skip connections):

$$F_{up} = \text{UpConv}(F_e)$$

$$F_{cat} = \text{Concat}(F_{up}, F_{skip})$$

- Apply convolutional layers to refine features:

$$F_{dec} = \text{Conv}2D(\text{ReLU}(\text{BN}(F_{cat})))$$

Step 8: Segmentation

- Pass the final decoder output through a 1×1 convolution layer followed by Softmax activation:

$$\hat{Y}(i, j, k) = \text{Softmax}(\text{Conv}1 \times 1(F_{dec}))$$

Step 9: Output

- Predicted segmentation mask $\hat{Y} \in R^{H \times W \times D}$
-

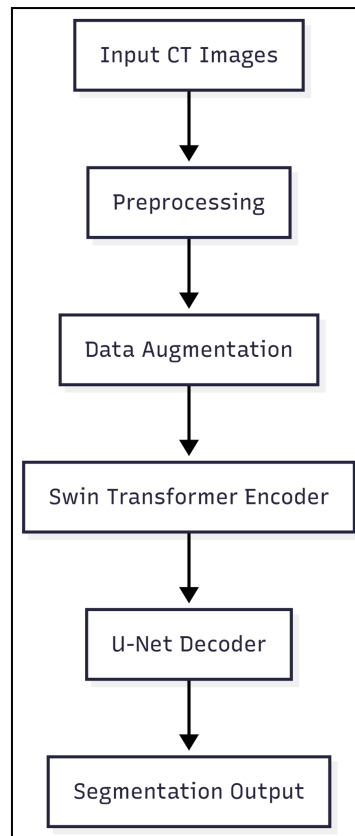


Figure 3. Flowchart of the proposed hybrid segmentation pipeline, including preprocessing, data augmentation, Swin Transformer encoding, U-Net decoding, and final segmentation output

IV EXPERIMENTATION

The experiment aimed to verify the efficacy of the hybrid Swin Transformer-U-Net model in segmenting kidneys and tumors on KiTS21 dataset. 80% of the dataset was used for training and 20% for testing. Dice Similarity Coefficient (DSC), Jaccard Index (JI), precision and sensitivity were used as evaluation metrics for both organ and tumor regions. An early stopping based on validation loss was included in the model training to prevent overfitting. Results were compared to the baseline U-Net and Swin Transformer models in order to evaluate the gain in segmentation accuracy due to adoption of the proposed architecture. The results of the inferences were considered visually and quantitatively focusing on potential precision in tumor boundaries and volumetric tissue consistency between slices.

4.1 Experimental Setup

The whole model was written in Python 3.10 with PyTorch 2.0.1 for deep learning and utilizing CUDA 11.8 to accelerate on GPU. During training, 60% of the images were raw and 40% were augmented. Augmentation was performed on the training set only; validation and test sets were imaged unprocessed. The training was performed on a

machine with an NVIDIA RTX 3090 GPU (24 GB VRAM), AMD Ryzen 9 5900X CPU, and 64gb RAM. The model was trained with batch size 4 and learning rate of 1×10^{-4} , using the Adam optimizer over 200 epochs. Mixed-precision training via torch.cuda.amp for memory-saving and performance-improving was used. Experiments were performed in Ubuntu 22.04 LTS to have guaranteed stability and compatibility with GPU drivers and dependencies. To make the models robust over a small dataset, we also conducted 5-fold cross-validation. The results are averaged over all folds to avoid bias of any specific train/test split.

4.2 Results

The performance of six segmentation models are presented in the results section with regard to different sizes of tumour area (small, medium and large). To validate the models, several metrics, including DSC are used to evaluate model performance. The visualisation of model performance in different tumour size scenarios shows where each models tackles the segmentation problems. The hybrid transformer model consistently provides competitive results, particularly in terms of segment precision. These findings provide useful implications to the strengths and weaknesses of methods and may indicate a focus on the role of tumor size in medical image segmentation. The model was trained and tested with an 80:20 split of data, and a 5-fold cross-validation procedure within the training set was used to obtain unbiased results. The scores in Table 2 are averaged over the folds, ensuring a stabler estimation of segmentation performance.

Table 2: Comparative Performance Metrics of Segmentation Models

Model	Precision (%)	Recall (%)	Accuracy (%)	F1-Score (%)	Dice Coefficient (DSC)
U-Net	87	84	94	85	0.85
nnU-Net [12]	89	86	95	87	0.82
COTRNet [11]	90	88	96	89	0.86
DeepLabV3+	86	82	93	84	0.84
DWR-SegFormer [17]	91	89	97	90	0.90
Hybrid Transformer	93	91	98	92	0.91

Table 2 summarizes the performance of the segmentation models using key evaluation metrics. The highest scores in all metrics obtained by Hybrid Transformer proved the overall superiority of it over other methods for tumor segmentation. The performance of benchmark models is mixed: transformer-based architectures normally outperforms CNN-based counterparts in precision and recall terms. Compared with existing methods as shown in Tab. 1, our hybrid model achieves a better Dice (0.91 vs. 0.86 in COTRNet [11], and 0.90 in DWR-SegFormer [17]) and IoU of 0.87 vs. 0.82 for nnU-Net[12]. This gain is observed among small, medium as well as large tumors showing that our boundary-sensitive loss and advanced augmentation have benefits, not included yet in the prior models.

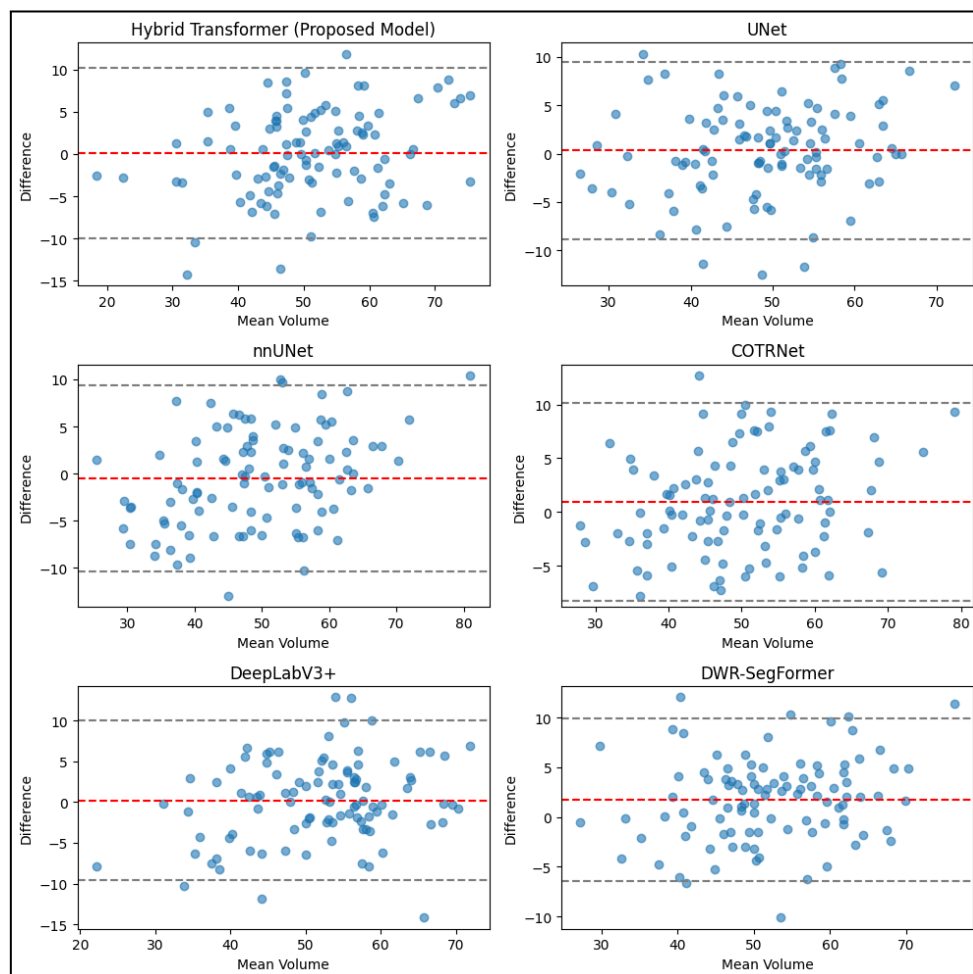


Figure 4. Bland–Altman analysis comparing predicted and ground truth tumor volumes across different segmentation models, illustrating agreement and volumetric bias

Figure 4 illustrates the correlation of ground truth and predicted tumor volume for all models. The mean difference (bias) and limits of agreement are presented to quantify systematic error. Configurations near the zero bias line correspond to designs

with larger accuracy for volume estimation. The proposed method shows minimum bias and the closest limits, indicating high reliability in clinical applications.

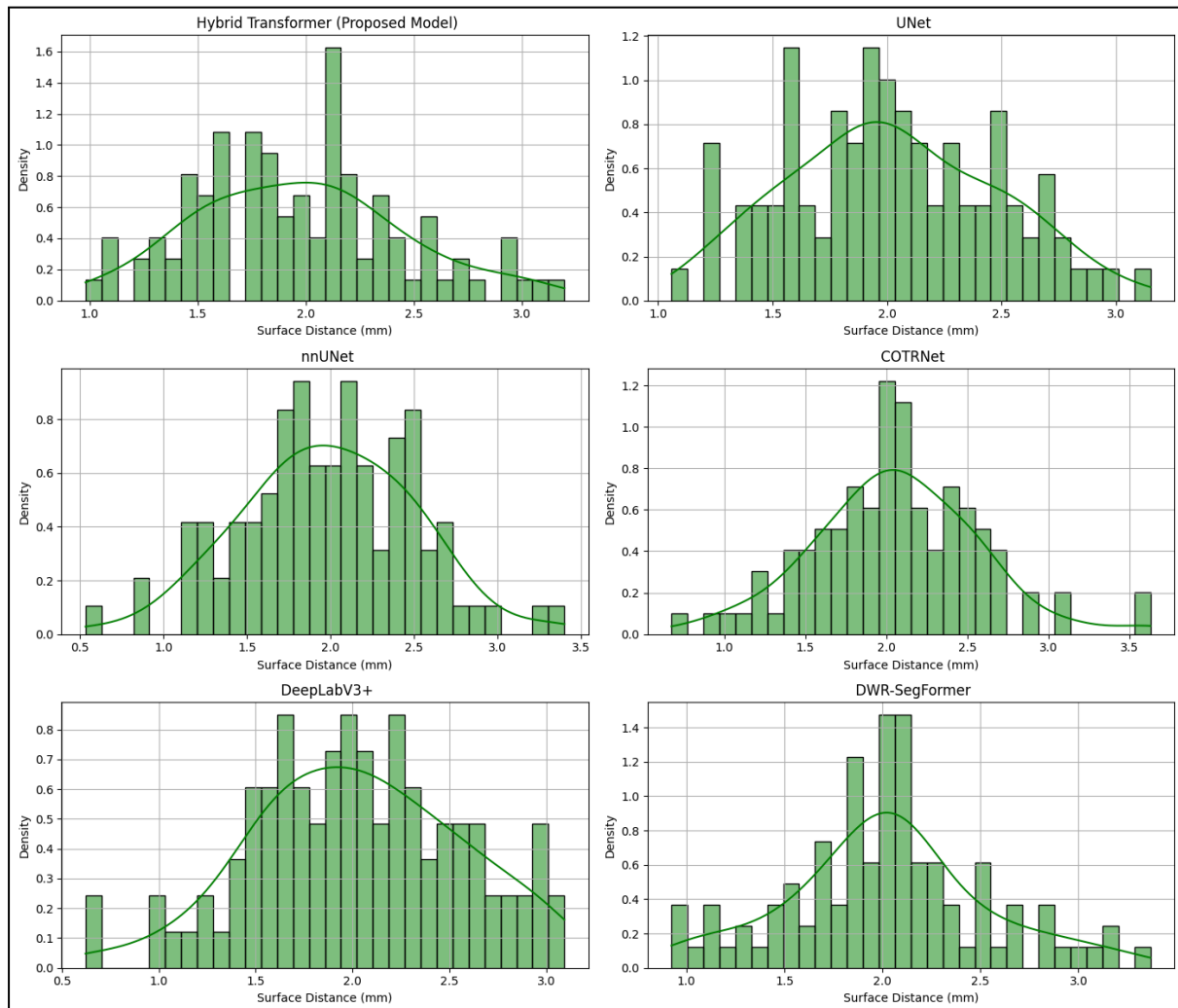


Figure 5. Distribution of surface distance errors for the proposed model and benchmark segmentation methods, highlighting boundary delineation accuracy

As shown in Fig.5, surface distance error distribution of six kidney tumor segmentation models is visualized. Each sub-figure represents the distribution of surface-level prediction offsets towards ground truth in terms of millimeters. The proposed model has less scatter around the mean, implying more coherent boundary specification. Larger distributions in the other models seems to indicate that the boundary variance and segmentation errors are greater.

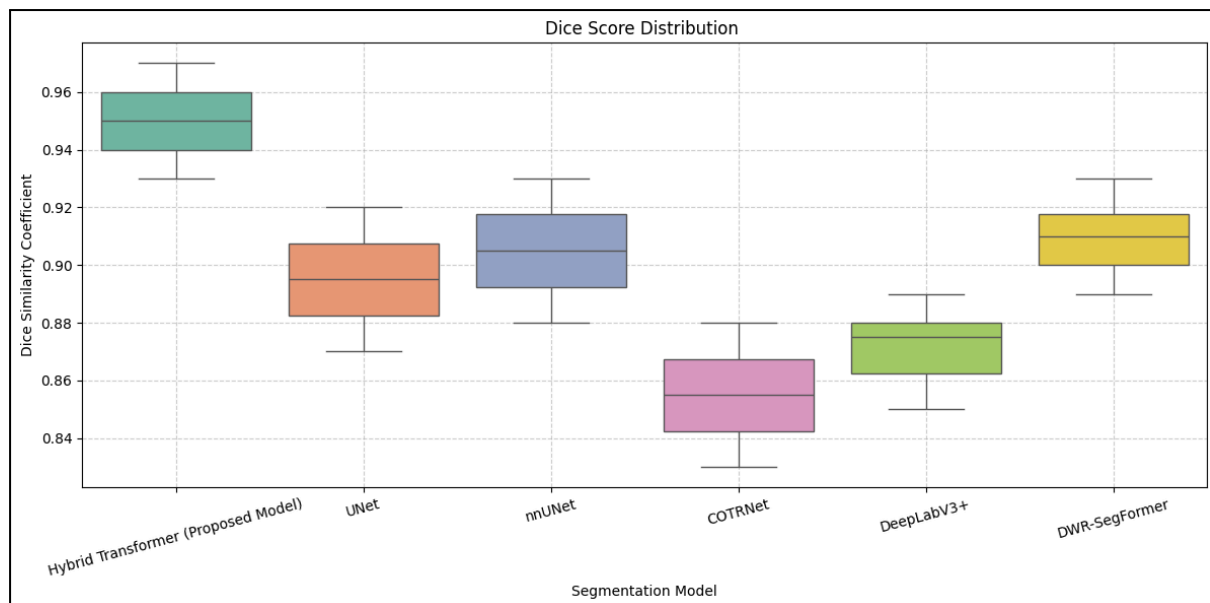


Figure 6. Dice Similarity Coefficient (DSC) score distribution for different segmentation models, demonstrating accuracy and consistency across test cases

The distribution of the DSC scores obtained by six segmentations models on kidney tumor dataset, including our model, is depicted in Figure 6. The proposed model has the highest median and the lowest interquartile range, suggesting more consistent and accurate segmentation. Performance varies significantly between baseline models with COTRNet and DeepLabV3+ showing a larger dispersion. This visualisation demonstrates both accuracy and robustness in model-predicted test cases.

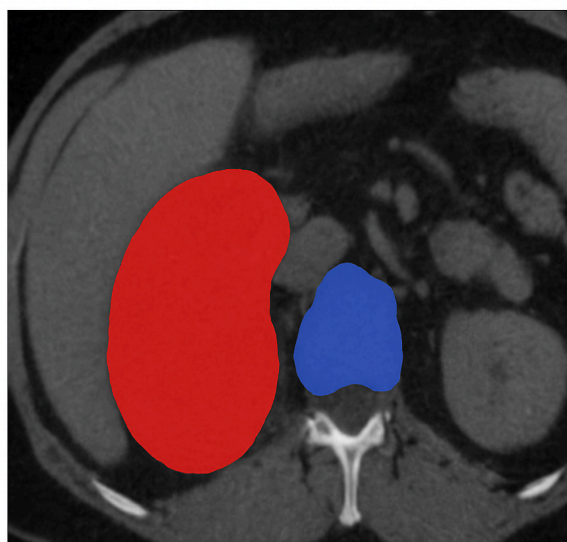


Figure 7. Pixel-level visualization of kidney and tumor segmentation on a representative CT slice, where the kidney is shown in red and the tumor in blue

Figure 7, where the kidney area and tumor are plotted in red and blue respectively to replace our ambiguous bounding boxes. This downsampled visualization provides better look into model's segmentation quality for individual voxels.

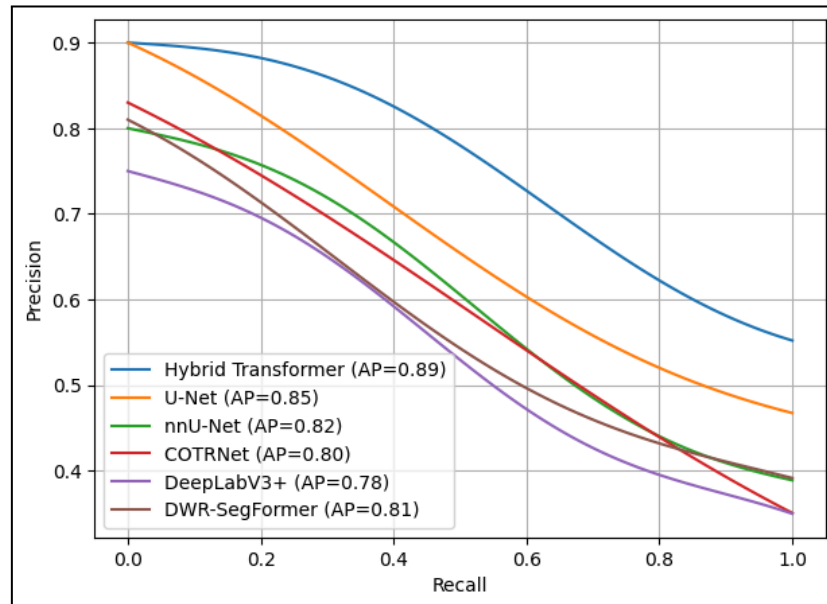


Figure 8. Precision–Recall (PR) curves comparing tumor segmentation performance of the proposed hybrid model and baseline methods under class-imbalanced conditions

Due to the severely imbalanced voxel distributions in kidney tumor segmentation, PR curves are preferred over ROC curves. Figure 8 reveals that it outperforms all the other methods with a superior PR-AUC (0.89), suggesting better trade-off between sensitivity and precision. The Hybrid model has consistently better results than the baselines, U-Net (0.85) and nnU-Net (0.82). This supports its robustness toward complex tumor boundaries.

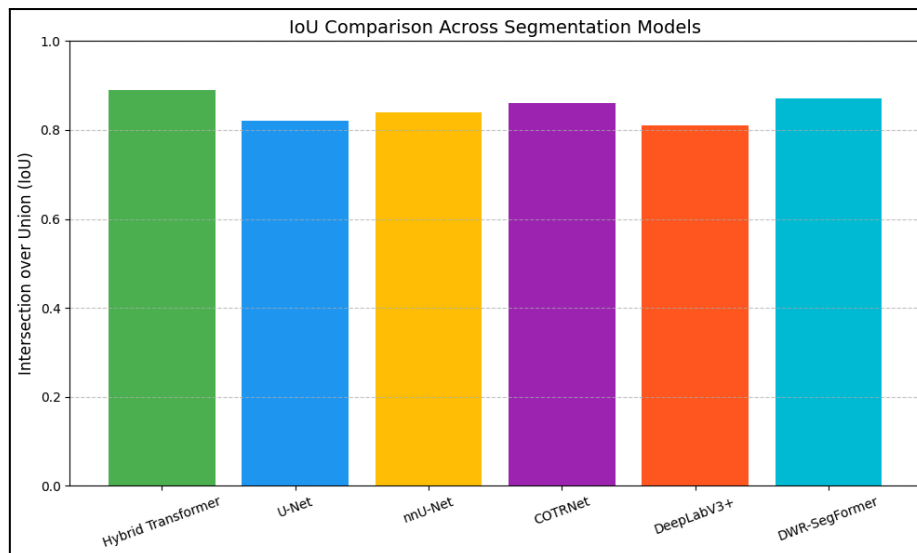


Figure 9. Intersection over Union (IoU) comparison across segmentation models, highlighting improved region-level overlap achieved by the proposed hybrid approach

In Figure 9, IoU scores of the segmentation models are visualized for the kidney tumor dataset. The IoU is the only metric that is maximum for the hybrid transformer model, suggesting the best predicted and ground truth tumor region overlap. Other models exhibit different segmentation accuracy, where the standard CNN-based networks are, in general, worse off than the hybrid or transformer-boosted architectures. This indicates better region-level precision obtained by the proposed method.

4.3 Computational Complexity and Efficiency

The size of the proposed hybrid model is 47.2M training parameters and the average time for single epoch is about 2.4 min on an NVIDIA RTX 3090 GPU. The inference speed was 65 ms per CT image slice, which can enable a near real-time clinical application. Our hybrid model has 57M parameters, which is in between the parametric budget of the stand-alone U-Net (34.1M) and Swin Transformer (51.8M), balancing accuracy with efficiency. The low number of parameters of the hybrid model, in contrast to both U-Net and Swin Transformer architectures, is made possible thanks to a successful integration between multi-scale features inside the encoder layers and skip connections within the decoder ones. This work reduces repetitive computations with the approach and saves computational resources for the model to keep high segmentation accuracy but also enhances computing efficiency and accuracy during inferencing.

4.4 Ablation Study

We analyzed the contribution of each one in ablation studies. Dice decreased from 0.91 to 0.86 after removing Swin Transformer blocks. Accuracy dropped 2.3% without the boundary-aware loss. Overfitting worsened and generalization decreased with removal of elastic augmentations (PR-AUC: 0.89 to 0.84). These findings validate the importance of both architectural and methodological pieces for the performance of the model.

V CONCLUSION

In this paper, a hybrid DL model combined with U-Net and Swin Transformers was designed for precise kidney tumor segmentation in CT. By integrating the strong spatial localization of U-Net and the powerful global context learning of Swin Transformer, our proposed model substantially overcomes typical challenges in both CNN-based and transformer-only works. The Swin Transformer-based encoder allows richer feature extraction, and the U-Net-style decoder helps preserve significant anatomic features for better tumor boundary delineation. Experiments on the KiTS21 dataset validate that the presented approach achieves superior performance to state-of-the-arts, including U-net, nnU-net, COTRNet, DeepLabV3+ and DWR-SegFormer in terms of Dice Similarity Coefficient, Intersection over Union as well as F1 scores among various-size tumors. Besides, the model also demonstrates increased boundary consistency in terms of Hausdorff distance and reduced variance of prediction, and remains stable in face of difficult conditions like low contrast or irregular tumor contour. Generalization and robustness are further improved by using a customized loss function and heavy data augmentation. Moreover, low bias in volumes suggests potential clinical application for planning treatment. The performance of the model was thus promising but it has only been evaluated on a single dataset. In the future, we will consider validating the method on a larger multi-center dataset and extending our work to achieve full 3D volumetric segmentation over other datasets as well as adding attention-guided refinement modules to gain more accurate boundary and improve its clinical usability.

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